

CST Third Semester Midterm Examination, September 2023

Signals and Systems (CS-2104)

Answer Q1 and any three questions from the rest.
Weightage of Q1 is double that of the rest.
Partial or incomplete answer to Q1 may be supplemented by answering all of the rest.
All part questions carry two marks each.

Full Marks: 30

Time: 120 minutes

- What is the energy of a periodic continuous signal over its specified time period?
 - How is the energy obtained from time domain information in case of discrete signal?
 - How can energy expression be derived when frequency domain information is available for a signal?
 - Check time invariance of the system whose output in time domain is the input multiplied by time.
 - Check linearity of a system whose output is the input added with some constant value.
 - Obtain impulse response of linear time invariant system.
- Obtain the Fourier coefficients of a square wave.
 - Explain how this can be extended to cover the frequency domain of an aperiodic rectangular pulse.
 - What frequencies are expected to be there in the output signal when the square wave is passed through an integrator?
- A second order continuous system is described by the differential equation:
 $\ddot{y} + 8\dot{y} + 15y = 3\dot{x} + x$
 - Obtain the transfer function $H(j\omega)$ of the system.
 - Obtain the output expression $y(t)$ for the input $x(t) = e^{-5t}u(t)$.
 - Sketch the input and tentative output waveforms.
- Discrete Fourier transform (DFT) of a signal is

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n}$$

- Obtain the DFT of the differentiation (difference) signal $x[n] - x[n-1]$.
- Hence obtain the DFT of the integration (accumulation) signal

$$y[n] = \sum_{m=-\infty}^{+n} x[m]$$

- Sketch the frequency domain for a discrete sequence $x[n] = a^n u[n]$ with $|a| < 1$.
- Discuss the basic concept of low pass filter design using RC circuit.
 - Explain the trade-off involved in low pass filter design.
 - What change is necessary to convert the low pass RC filter to high pass filter?